

Facial Emotion Recognition using AlexNet

Based on the AffectNet Research Paper



Research Internship under Prof. Anirban Dasgupta

Presented by :

Sowmya Sri

B.Tech 4th Year,

Electronics & Communication Engineering

Indian Institute of Technology Guwahati

Little History about AffectNet :

Facial expressions provide valuable cues about human emotions and are widely studied in **affective computing**.

AffectNet is a large-scale facial expression database designed to support robust emotion recognition in real-world conditions

MODELS TO MEASURE EMOTION

- **Categorical**
Emotion represented as discrete labels(Happy, Sad, etc..)
- **Dimensional**
Emotion represented as continuous values(Valence & Arousal)
- **FACS (Facial Action Coding System)**
Represents facial muscle movements using **Action Units (AUs)**

Why Existing Databases Failed?

- **Too small**
Only a few hundred to thousand images, which was insufficient for training deep learning models.
- **Lab-controlled and posed**
Expressions were captured in controlled environments and did not represent natural human emotions.
- **Single emotion model**
Most datasets supported either categorical emotions or dimensional emotions, but not both.
- **Limited diversity**
Included fewer subjects with limited variations in age, ethnicity, head pose, lighting, and background, resulting in poor generalization.

AffectNet Database:

- **Massive scale**
Over 1 million facial images collected from Google, Bing, and Yahoo using 1,250 emotion-related keywords in 6 languages.
- **Real-world images**
Contains natural, unposed facial expressions with diverse lighting conditions, backgrounds, and viewpoints.
- **Supports both emotion models**
Provides categorical emotions (8 classes) as well as dimensional annotations (Valence and Arousal) in a single dataset.
- **Expert annotations**
More than 450,000 images were manually labeled by 12 trained human experts, ensuring high-quality annotations.

This work focuses on implementing the **categorical emotion recognition** approach of the AffectNet paper using the **AlexNet architecture**, trained from scratch on the **FER2013** dataset and systematically evaluates different optimization strategies, including standard training, data augmentation, and class-weighted loss, to improve emotion classification performance. The trained model is analyzed using standard evaluation metrics and confusion matrix-based performance analysis.

Dataset Used : **FER2013**(7 classes, all grayscale images)

Training Set(28709 total images)

Emotion	Images
Angry	3,995
Disgust	436
Fear	4,097
Happy	7,215
Neutral	4,965
Sad	4,830
Surprise	3,171

Testing Set(7178 total images)

Emotion	Images
Angry	958
Disgust	111
Fear	1,024
Happy	1,774
Neutral	1,233
Sad	1,247
Surprise	831

Architecture used : AlexNet

- AlexNet is a deep Convolutional Neural Network (CNN) consisting of
 - 5 convolution layers,
 - 3 max-pooling layers,
 - 2 fully connected layers and 1 SoftMax layer.
 - It is used for image classification task.

Convolution layer



Extracts patterns in images.

In each layer, there are multiple filters, each filter giving one activation map or feature map

Max Pooling layer



Removes unwanted information, reducing dimensionality.

Fully Connected layers with
ReLU activation function
(Multilayer perceptrons)



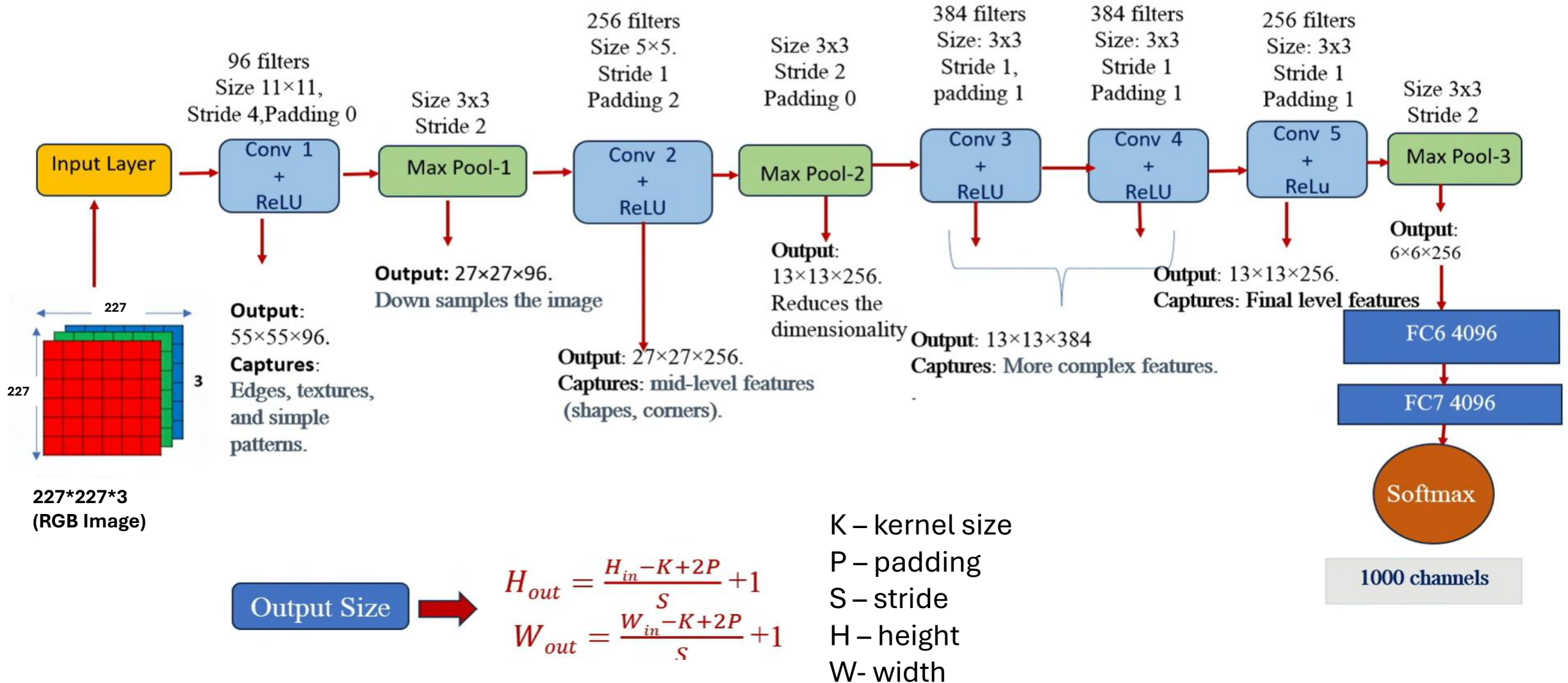
Responsible for learning the high-level representations and making predictions

Softmax layer



Classifies the images into 1000 classes

Layer-wise Working of AlexNet :



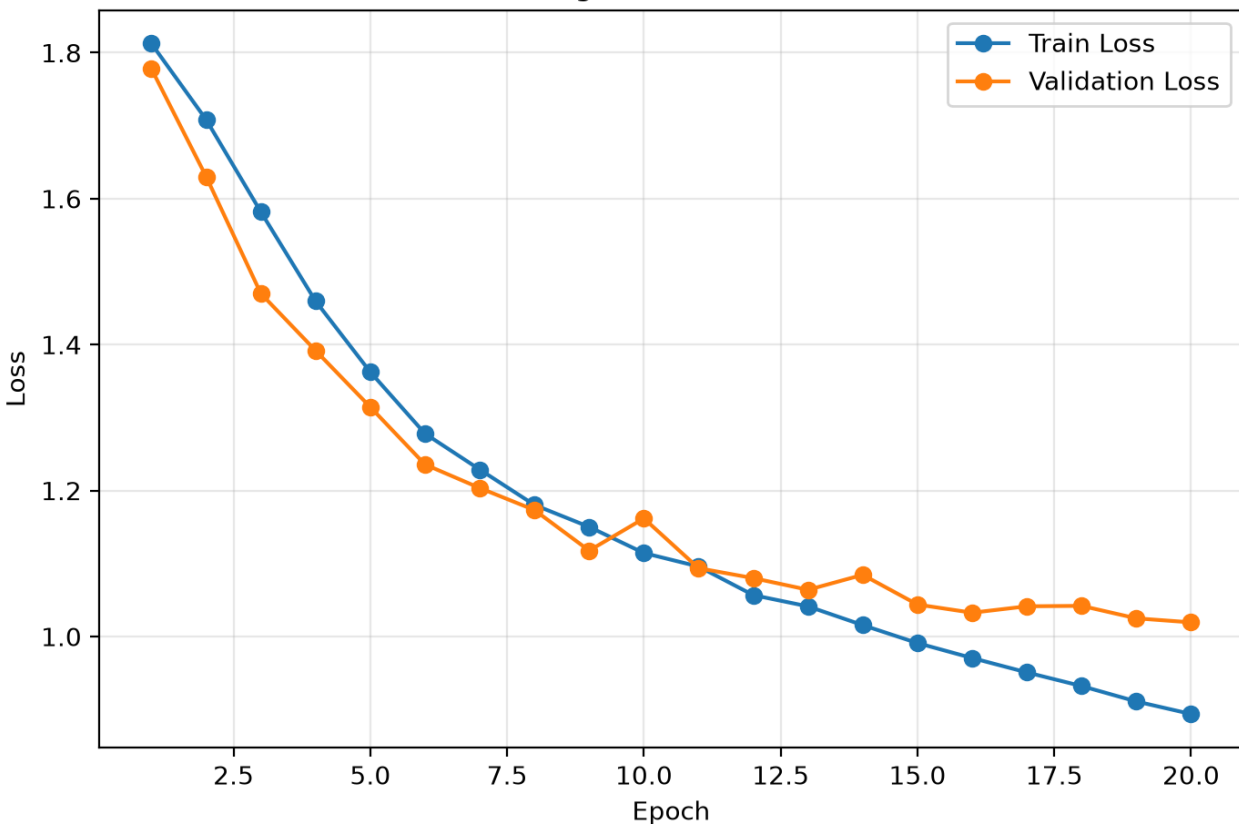
Implementation details in brief:

The FER2013 dataset was used to train and evaluate the AlexNet model for seven-class facial emotion recognition.

- AlexNet expects a 3-channel RGB input, so **each grayscale image** was replicated across **three channels**.
- All **images were resized to 227×227 pixels**, normalized, and split into training, validation, and testing sets.
- Data augmentation techniques such as **Random Horizontal Flip** and **Random Rotation** were applied to improve generalization and reduce overfitting.
- Two training strategies were used : **Imbalanced and weighted loss**, performance was evaluated using **Accuracy, Precision, Recall, F1-score, Confusion Matrix, and Classification Report**.
- **CrossEntropyLoss** was used, for imbalanced approach, without assigning class weights and applying class weights for weighted-loss approach.
- The model was trained for **20 epochs** using **Stochastic Gradient Descent (SGD)** with a **learning rate of 0.01, momentum of 0.9, and weight decay of 0.0005**.
- The best-performing model was selected based on the **highest validation accuracy**.

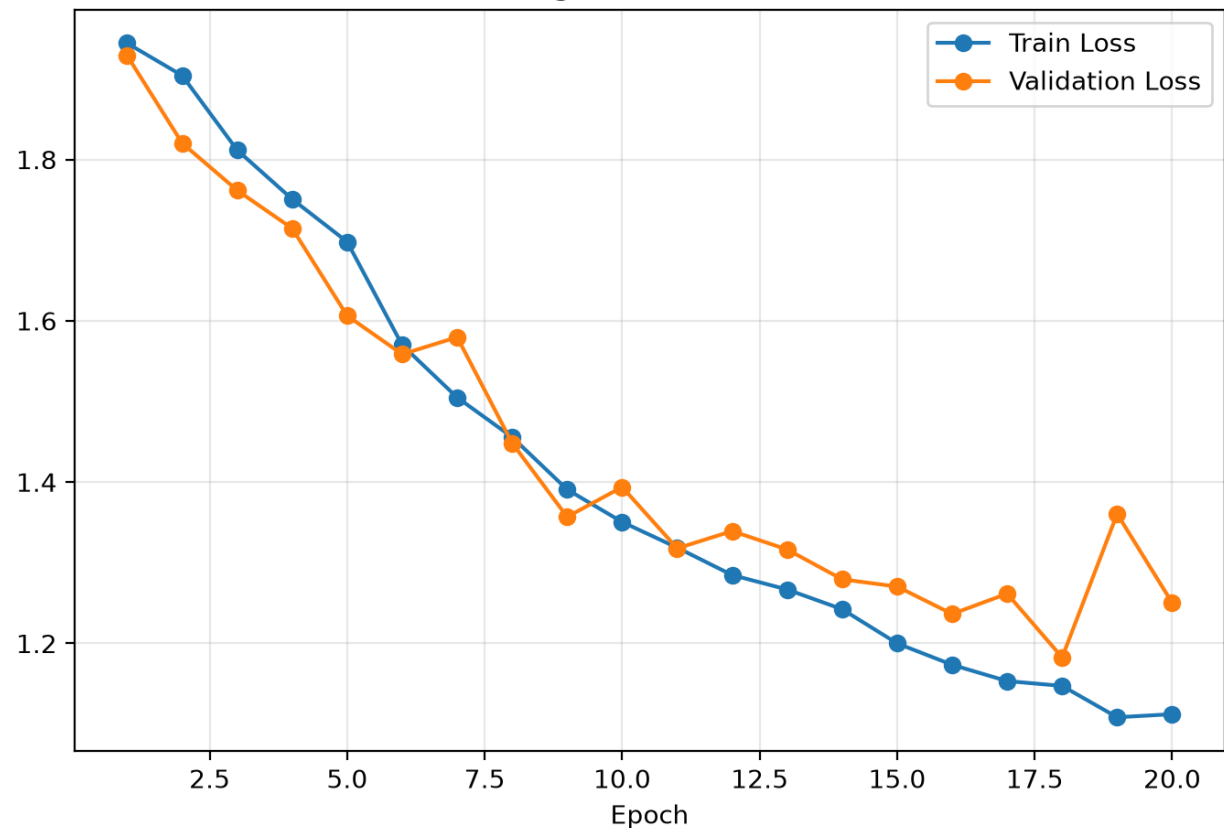
Training and Validation Loss Comparison

Training vs Validation Loss



Imbalanced Training

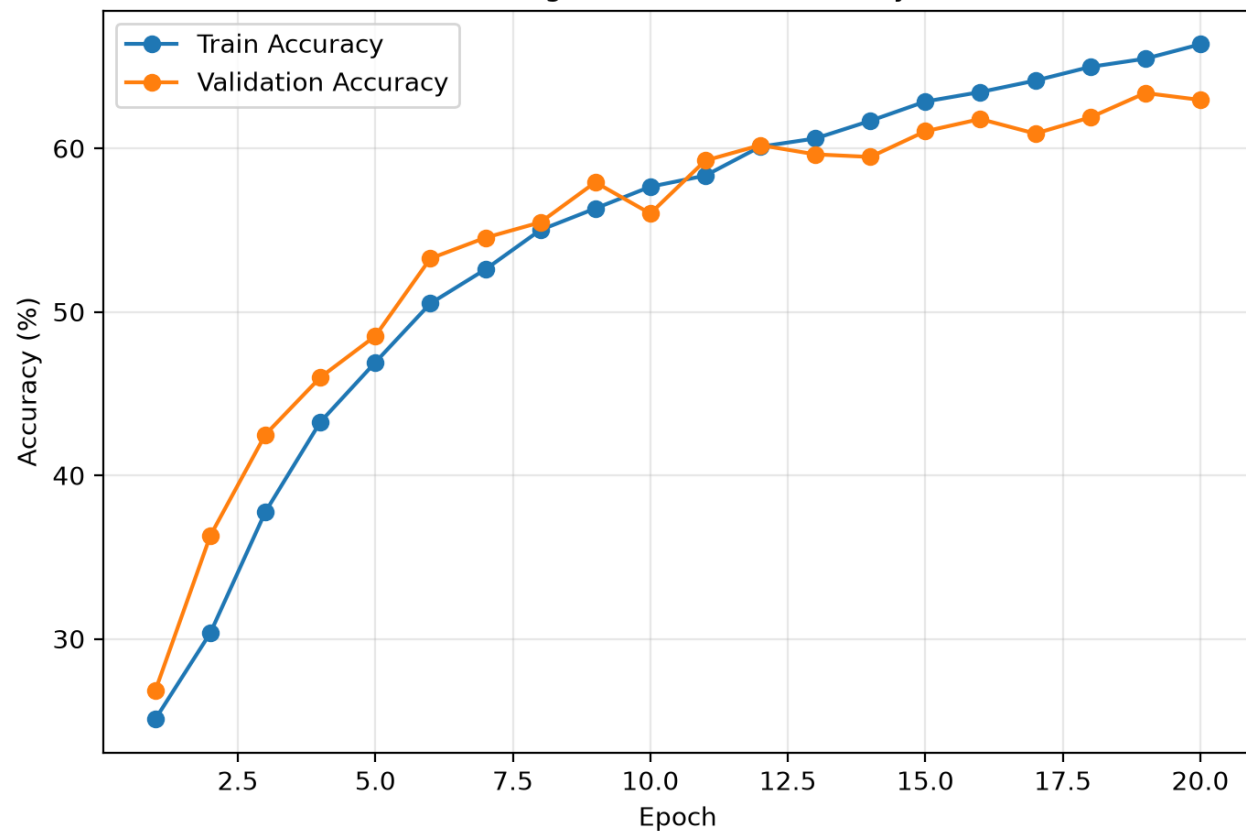
Training vs Validation Loss



Weighted-Loss Training

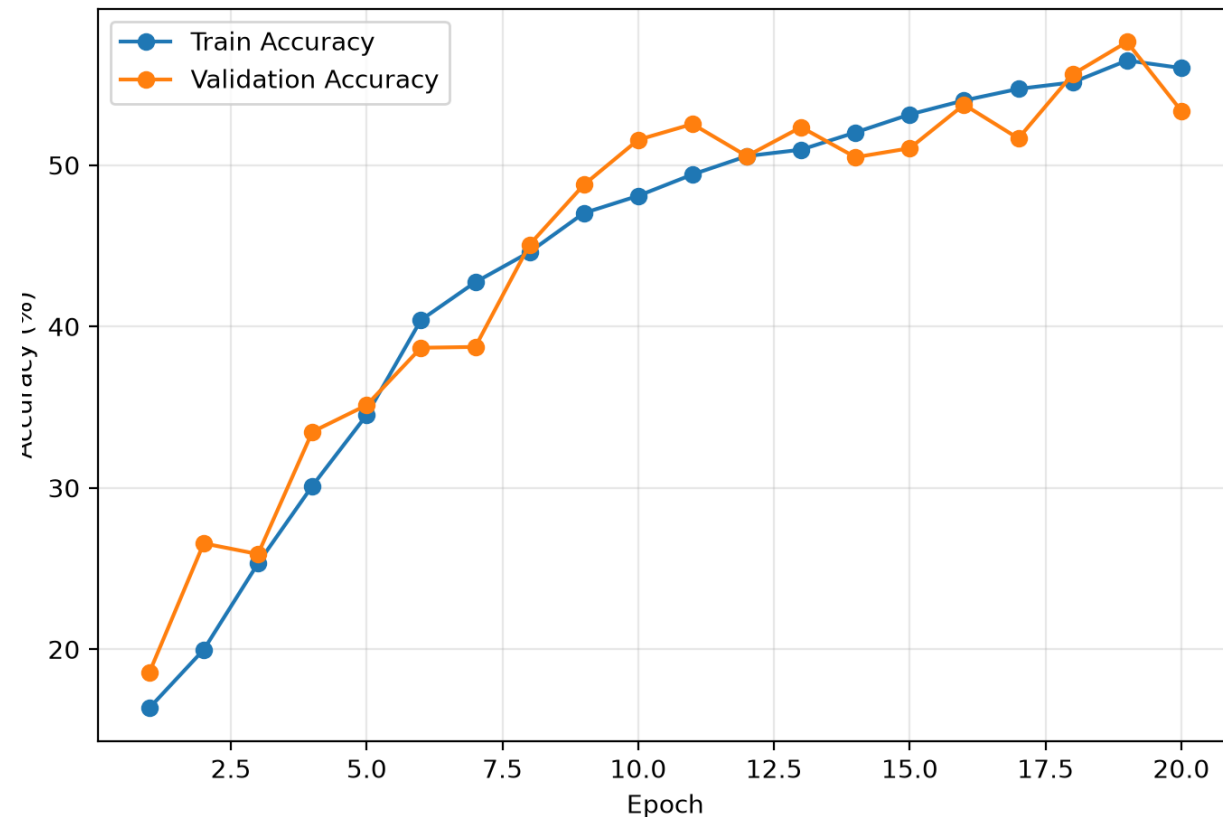
Training and Validation Accuracy Comparison

Training vs Validation Accuracy



Imbalanced Training

Training vs Validation Accuracy



Weighted-Loss Training

Comparison in F1 score in both approaches :

	Imbalanced				Weighted-Loss			
	Top 1		Top 2		Top1		Top2	
classes	orig	norm	orig	norm	orig	norm	orig	norm
Angry	0.54	0.51	0.73	0.69	0.47	0.48	0.67	0.66
Disgust	0.41	0.46	0.57	0.60	0.49	0.59	0.62	0.69
Fear	0.37	0.41	0.72	0.70	0.35	0.40	0.68	0.74
Happy	0.84	0.82	0.91	0.89	0.79	0.78	0.88	0.87
Neutral	0.60	0.57	0.78	0.74	0.56	0.53	0.74	0.71
Sad	0.52	0.51	0.79	0.72	0.49	0.47	0.74	0.69
Surprise	0.76	0.78	0.86	0.89	0.72	0.74	0.85	0.88

- Orig and Norm stand for Original and skew-Normalized

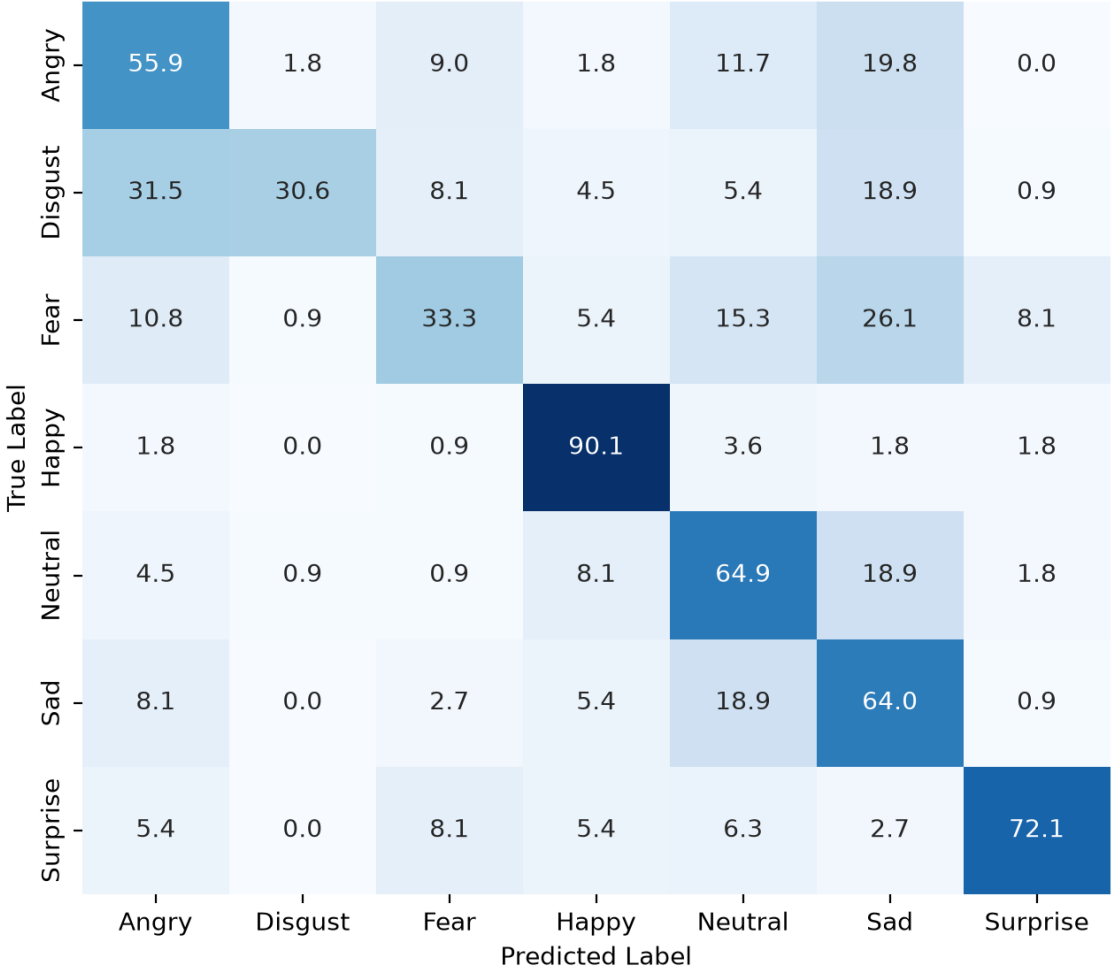
Observation :

The weighted-loss method improved the F1-score of minority class **Disgust**. However, it slightly reduced the performance of majority classes such as **Happy, Neutral, Sad, Fear, Angry, and Surprise**. As a result, the overall weighted F1-score decreased because the gains in minority class performance were outweighed by the decline in the more frequent classes, which contribute more to the overall metric.

	Imbalanced		Weighted-loss	
Total	Orig	Norm	Orig	Norm
F1-score	0.62	0.58	0.58	0.57

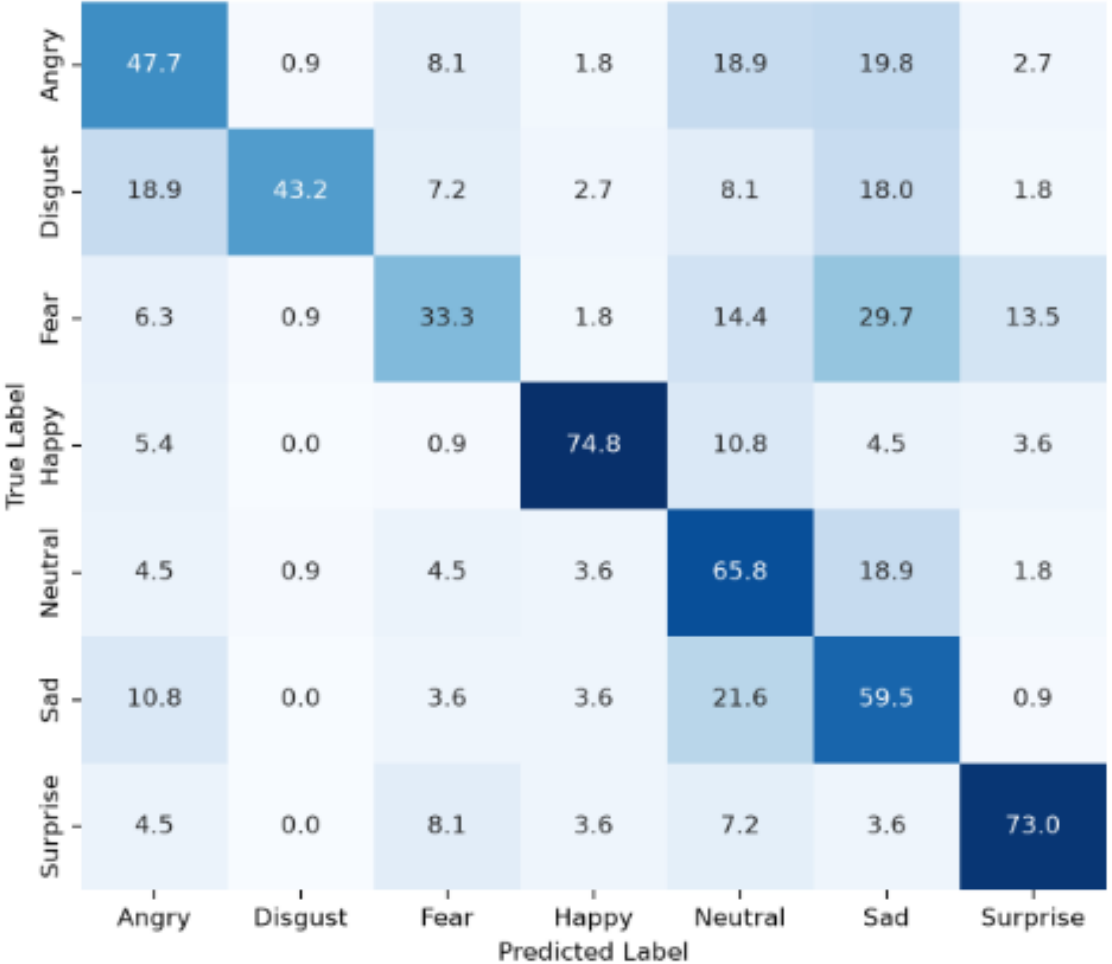
Imbalanced Approach

Confusion Matrix (%)



Weighted-loss Approach

Confusion Matrix (%)



Compared to the baseline Imbalanced, **weighted loss successfully improves the minority Disgust class** while slightly decreasing the classification performance of several majority classes. **This trade-off explains the reduction in the overall weighted F1-score.**

Evaluation Metrics :

	Imbalanced		Weightedloss	
	Orig	Norm	Orig	Norm
Accuracy	62.76	58.69	57.59	56.76
F1-score	0.62	0.58	0.58	0.57
Alpha	0.55	0.52	0.49	0.49
Kappa	0.55	0.52	0.49	0.50
AUC	0.90	0.90	0.87	0.88
AUCPR	0.68	0.67	0.62	0.62

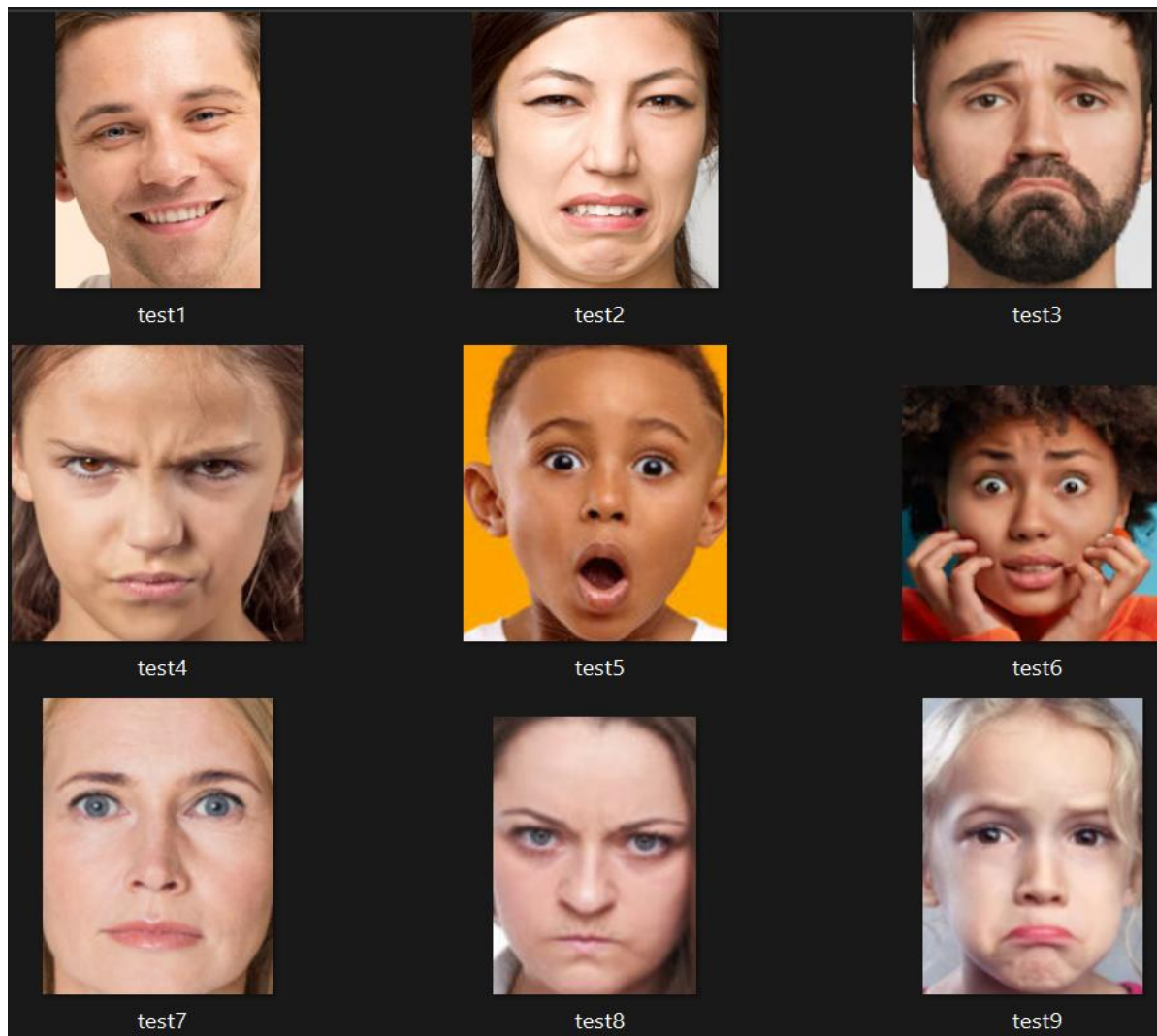
Top-1 Accuracy: 62.76% (Imbalanced-orig), 57.59% (Weighted Loss-orig)

Top-2 Accuracy: 80.64% (Imbalanced-orig), 76.66% (Weighted Loss-orig)

The **Imbalanced (Original)** approach consistently outperformed the Weighted Loss approach across all major evaluation metrics. Hence, it was selected as the **best-performing model** for the FER2013 dataset.

Emotion Prediction on Unseen Images :

Both trained AlexNet models were tested on unseen facial images to evaluate their real-world prediction capability. While both models correctly identified most prominent emotions, differences were observed for subtle expressions such as Neutral and Sad.



Imbalanced
Model →

```
test1.png      : happy
test2.png      : neutral
test3.png      : sad
test4.png      : angry
test5.png      : surprise
test6.png      : fear
test7.png      : neutral
test8.png      : angry
test9.png      : neutral
```

Weighted-Loss
Model →

```
test1.png      : happy
test2.png      : happy
test3.png      : neutral
test4.png      : angry
test5.png      : surprise
test6.png      : fear
test7.png      : neutral
test8.png      : angry
test9.png      : neutral
```

Conclusion :

- Implemented an **AlexNet-based categorical facial emotion** recognition system using the **FER2013** dataset and **PyTorch**.
- Compared two training strategies: **Imbalanced Training and Weighted Loss**.
- The **Imbalanced model** achieved the best performance (**62.76% Top-1, 80.64% Top-2 Accuracy, F1-score: 0.62**).
- **Weighted Loss improved minority-class** recognition but reduced the overall classification performance.
- The **trained model was further validated on unseen facial images**, demonstrating its ability to predict emotions in real-world scenarios.

Future Scope : the proposed improvements can further improve accuracy and real-world applicability.

- **Focal Loss:** Replace Weighted Cross-Entropy with **Focal Loss** to focus training on difficult samples and improve recognition of minority classes, while maintaining overall accuracy.
- **Deeper CNN Models:** Use **ResNet-50** or **EfficientNet**, which extract richer features and generally outperform AlexNet on facial emotion recognition tasks.
- **Transfer Learning:** Fine-tune **ImageNet pre-trained models** to achieve faster convergence and higher accuracy than training from scratch.
- **Face Detection & Alignment:** Detect and align faces before classification to remove background noise and improve prediction accuracy.
- **Larger & Better Datasets:** Train and evaluate on **AffectNet** or **RAF-DB**, which provide more diverse, higher quality facial expressions than FER2013.